

```
In [1]: # Name - Rohan Pise
        # Rollno. - 01
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In [3]: import tensorflow as tf
        from tensorflow import keras
        import matplotlib.pyplot as plt
        import numpy as np

        # Load MNIST dataset
        (X_train, y_train), (X_test, y_test) = keras.datasets.mnist.load_data()

        # Normalize the data (scale pixel values to 0-1)
        X_train, X_test = X_train / 255.0, X_test / 255.0

        # Build a simple CNN model
        model = keras.Sequential([
            keras.layers.Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)),
            keras.layers.MaxPooling2D(2,2),
            keras.layers.Conv2D(64, (3,3), activation='relu'),
            keras.layers.MaxPooling2D(2,2),
            keras.layers.Flatten(),
            keras.layers.Dense(128, activation='relu'),
            keras.layers.Dense(10, activation='softmax') # 10 classes (digits 0-9)
        ])

        # Compile the model
        model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['a

        # Train the model
        model.fit(X_train, y_train, epochs=5, validation_data=(X_test, y_test))

        # Evaluate the model
        test_loss, test_accuracy = model.evaluate(X_test, y_test)
        print("Test Accuracy:", test_accuracy)

        # Predict a random sample
        index = np.random.randint(0, len(X_test))
        plt.imshow(X_test[index], cmap='gray')
        plt.title(f"True Label: {y_test[index]}")
        plt.show()

        prediction = np.argmax(model.predict(X_test[index].reshape(1, 28, 28, 1)))
        print("Predicted Label:", prediction)
```

C:\Users\vidhi\anaconda3\Lib\site-packages\keras\src\layers\convolutional\base_conv.py:107: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Epoch 1/5

1875/1875 ————— **19s** 9ms/step - accuracy: 0.9103 - loss: 0.2914 - val_
accuracy: 0.9845 - val_loss: 0.0452

Epoch 2/5

1875/1875 ————— **20s** 11ms/step - accuracy: 0.9861 - loss: 0.0456 - val_
_accuracy: 0.9893 - val_loss: 0.0346

Epoch 3/5

1875/1875 ————— **20s** 11ms/step - accuracy: 0.9904 - loss: 0.0305 - val_
_accuracy: 0.9888 - val_loss: 0.0340

Epoch 4/5

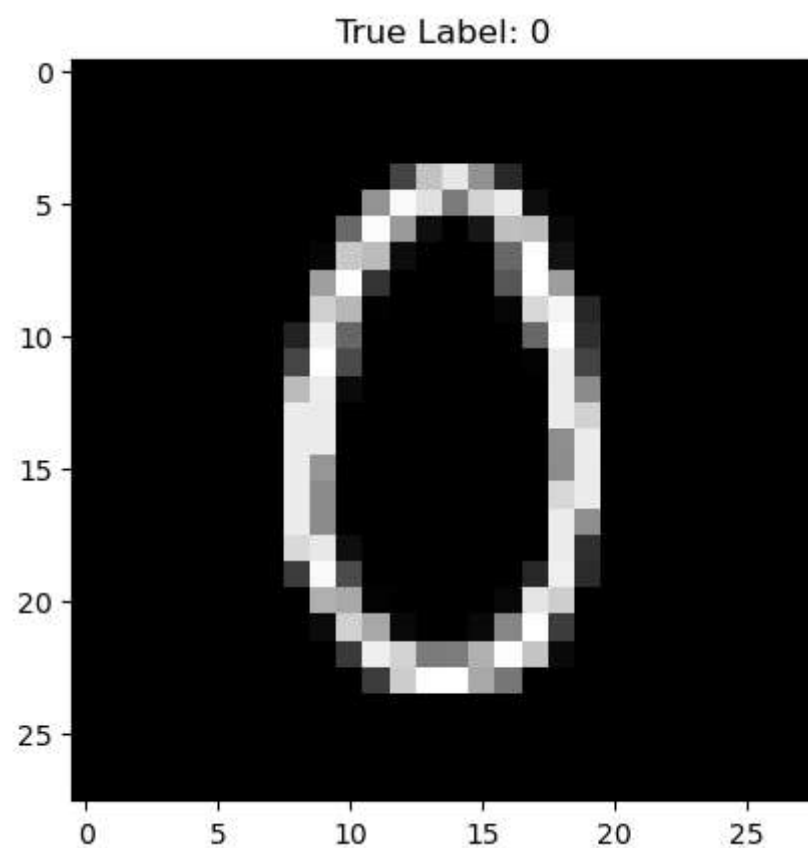
1875/1875 ————— **20s** 11ms/step - accuracy: 0.9935 - loss: 0.0208 - val_
_accuracy: 0.9906 - val_loss: 0.0282

Epoch 5/5

1875/1875 ————— **21s** 11ms/step - accuracy: 0.9953 - loss: 0.0146 - val_
_accuracy: 0.9913 - val_loss: 0.0276

313/313 ————— **2s** 5ms/step - accuracy: 0.9887 - loss: 0.0354

Test Accuracy: 0.9912999868392944



1/1 ————— **0s** 157ms/step

Predicted Label: 0

In []: